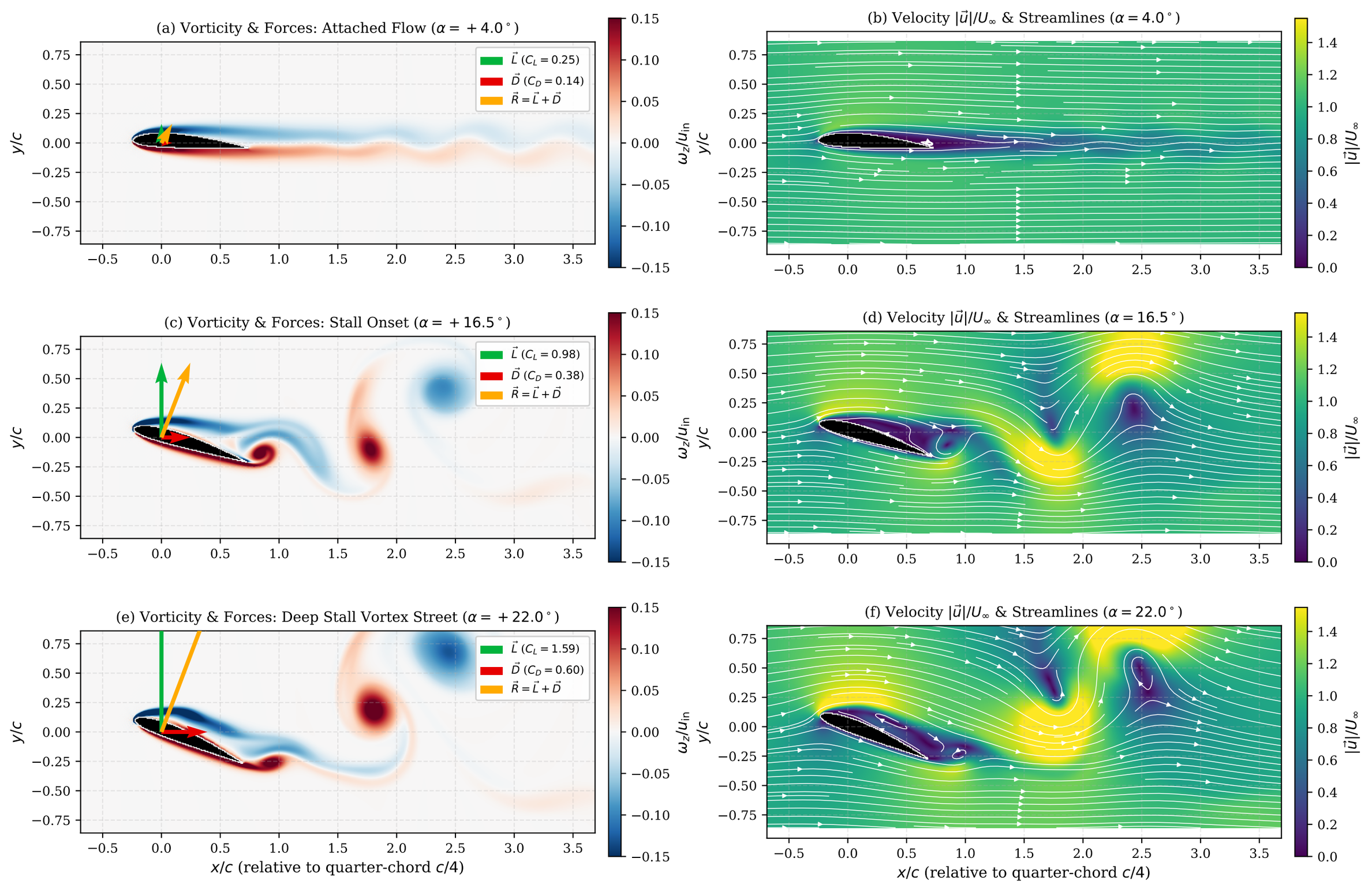

(a) Vorticity & Forces: Attached Flow ($\alpha = +4.0^\circ$)
$\vec{L}$ ($C_L = 0.25$)
$\vec{D}$ ($C_D = 0.14$)
$\vec{R} = \vec{L} + \vec{D}$
(b) Velocity $|\vec{u}|/U_\infty$ & Streamlines ($\alpha = 4.0^\circ$)
(c) Vorticity & Forces: Stall Onset ($\alpha = +16.5^\circ$)
$\vec{L}$ ($C_L = 0.98$)
$\vec{D}$ ($C_D = 0.38$)
$\vec{R} = \vec{L} + \vec{D}$
(d) Velocity $|\vec{u}|/U_\infty$ & Streamlines ($\alpha = 16.5^\circ$)
(e) Vorticity & Forces: Deep Stall Vortex Street ($\alpha = +22.0^\circ$)
$\vec{L}$ ($C_L = 1.59$)
$\vec{D}$ ($C_D = 0.60$)
$\vec{R} = \vec{L} + \vec{D}$
(f) Velocity $|\vec{u}|/U_\infty$ & Streamlines ($\alpha = 22.0^\circ$)
y/c
x/c (relative to quarter-chord $c/4$)
ω_z/u_{in}
$|\vec{u}|/U_\infty$